

Here is a structured summary of **Orienting Yourself: The Use of Coordinates**, covering how coordinate systems are used to identify positions, navigate, and read maps effectively.

1. What Are Coordinates?

- **Definition:** A set of numbers (or letters and numbers) used to specify the precise location of a point or place on a map, surface, or space.

- **Purpose:** To eliminate ambiguity when describing a location—instead of saying "near the center," coordinates give an exact mathematical position.

2. Geographic Coordinate System (Earth's Grid)

When orienting yourself on the globe, location is determined using an imaginary grid formed by two sets of lines: **Latitude** and **Longitude**.

Latitude and longitude diagram, vector educational poster. Source: Shutterstock
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A. Lines of Latitude (Parallels)

- **Orientation:** Run horizontally (East to West), but measure distance **North or South** of the Equator.
- **Reference Line:** The **Equator (0°)** divides Earth into the Northern and Southern Hemispheres.
- **Range:** 0° at the Equator to 90° N (North Pole) and 90° S (South Pole).

B. Lines of Longitude (Meridians)

- **Orientation:** Run vertically (North to South, pole to pole), but measure distance **East or West** of the Prime Meridian.
- **Reference Line:** The **Prime Meridian (0°)**, passing through Greenwich, England.
- **Range:** 0° to 180° E and 180° W.

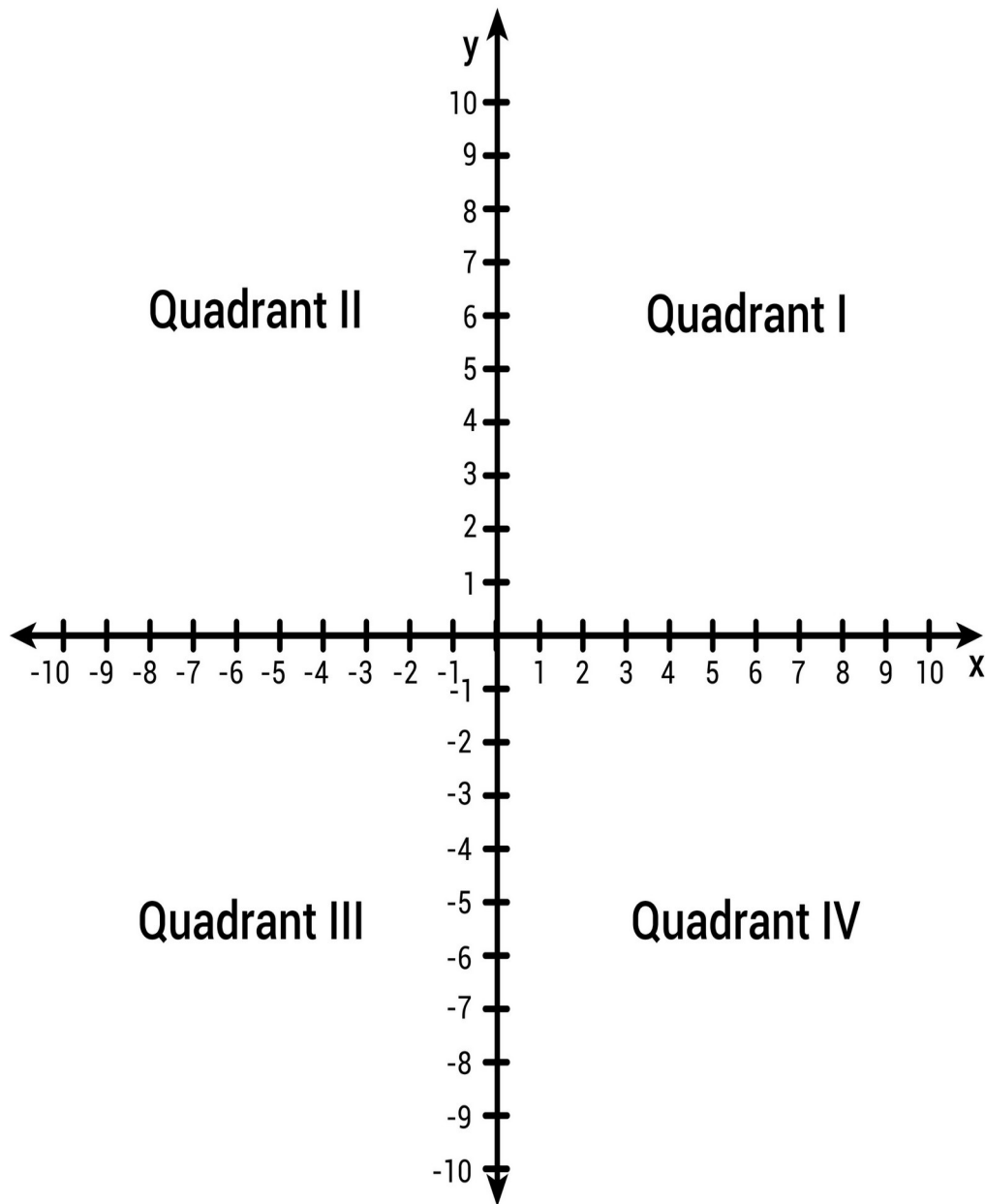
How to Write Geographic Coordinates: > Always state **Latitude first**, then **Longitude**.

- *Example:* 27.1751° N, 78.0421° E (Location of the Taj Mahal).

3. Cartesian Coordinate System (2D Grid Maps)

On flat maps, graph sheets, or city grids, locations are identified using the **Cartesian coordinate plane**:

Cartesian Coordinate System



Source: Shutterstock

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- **Horizontal Axis (X-axis):** Measures left-to-right (Easting).
- **Vertical Axis (Y-axis):** Measures bottom-to-top (Northing).
- **Origin (0,0):** The starting reference point where the axes cross.

- **Ordered Pair (x,y):** * x tells you how many units to move horizontally.
 - y tells you how many units to move vertically.

4. Grid References on Local Maps

In topographic maps and military navigation, a simplified grid system is often used:

Type	Format	Description
4-Figure Grid Reference	3245	Identifies a whole grid square (Grid Easting 32, Northing 45).
6-Figure Grid Reference	324458	Pinpoints an exact location <i>within</i> a grid square for higher precision.

5. Summary Key Takeaways

1. **Grid Systems:** Orienting yourself relies on a fixed reference grid.
2. **Two Inputs Needed:** Any point on a 2D surface requires two values (Latitude/Longitude, X/Y, or Easting/Northing).
3. **Cardinal Directions:** Coordinates are tied to orientation directions—North, South, East, and West.

THANK YOU 

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